# ARJUNA JEE 2027

Lecture – 3 | Physics

# Centre of mass & System of particles

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# Topics *to be covered*

Revised ~ 81%

A COM of Continuous System

B Combinational Problems

C Questions

D



## QUESTION 1

HW

A rod of length $L$ has non-uniform linear mass density given by $\lambda(x) = a + b\left(\frac{x}{L}\right)^2$, where $a$ and $b$ are constants and $0 \le x \le L$. The value of location of centre of mass of the rod is at :

**A** $\frac{4}{3}\left(\frac{a+b}{2a+3b}\right)L$

**B** $\frac{3}{4}\left(\frac{2a+b}{3a+b}\right)L$

**C** $\frac{3}{2}\left(\frac{2a+b}{3a+b}\right)L$

**D** $\frac{3}{2}\left(\frac{a+b}{2a+b}\right)L$




$\lambda = a + \frac{bx^2}{L^2}$

$X_c = \frac{\int x\,dm}{\int dm}$

$dm = \lambda dx$

$dm = \left(a + \frac{bx^2}{L^2}\right)dx$

$$X_c = \frac{\int_0^L x\left(a+\frac{bx^2}{L^2}\right)dx}{\int_0^L \left(a+\frac{bx^2}{L^2}\right)dx} \quad \text{Variable}$$

$$= \frac{\int_0^L ax\,dx + \int_0^L \frac{bx^3}{L^2}dx}{\int_0^L a\,dx + \int_0^L \frac{bx^2}{L^2}dx}$$

$$= \frac{a\left[\frac{x^2}{2}\right]_0^L + \frac{b}{L^2}\left[\frac{x^4}{4}\right]_0^L}{a\left[x\right]_0^L + \frac{b}{L^2}\left[\frac{x^3}{3}\right]_0^L} = \frac{\frac{aL^2}{2} + \frac{bL^{\cancel{4}2}}{4\cancel{L^2}}}{aL + \frac{bL^3}{3\cancel{L^2}}}$$

$$= L\left[\frac{\frac{a}{2}+\frac{b}{4}}{a+\frac{b}{3}}\right] = L\left[\frac{\frac{2a+b}{4}}{\frac{3a+b}{3}}\right]$$

$$\boxed{X_c = \frac{3L}{4}\left[\frac{2a+b}{3a+b}\right]}$$

**QUESTION 2**

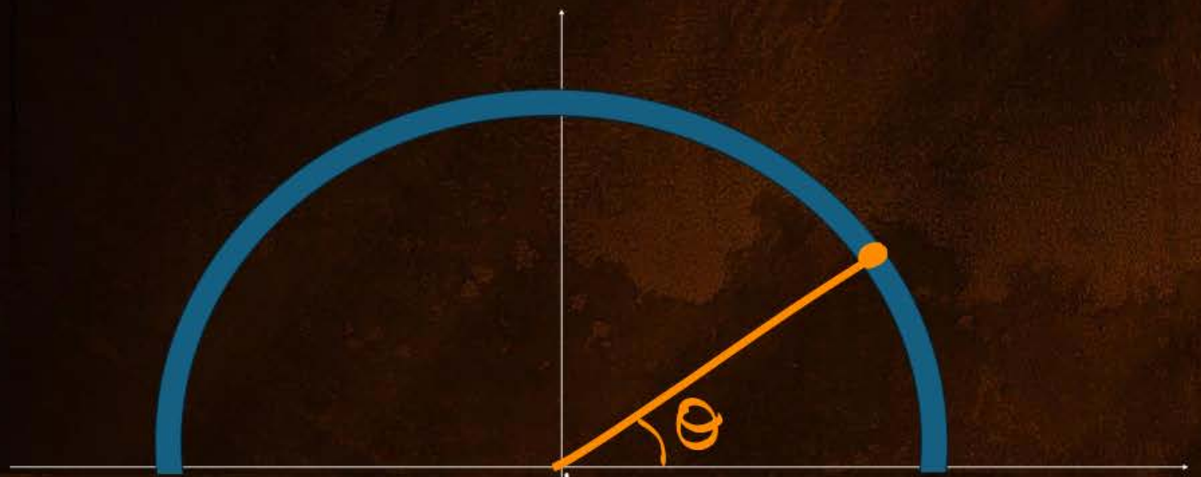
HW

$\lambda = \lambda_0 \sin\theta$.

(Non uniform Semi Ring).

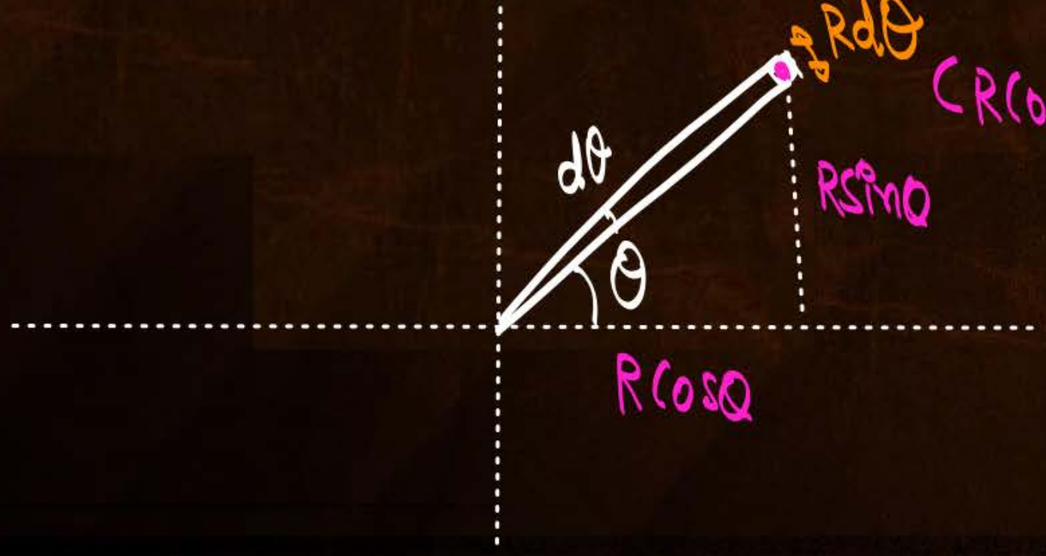
$$Y_c = \frac{\int y\, dm}{\int dm}$$

$y$ → coordinate of COM of element




$dm = \lambda dx$

$= (\lambda_0 \sin\theta) R d\theta$




(RCosθ, RSinθ).

$$Y_c = \frac{\int (R\sin\theta)(\lambda_0 R \sin\theta\, d\theta)}{\int \lambda_0 R \sin\theta\, d\theta}$$

$$= \frac{\lambda_0 R^2 \int_0^{\pi} \sin^2\theta\, d\theta}{\lambda_0 R \int_0^{\pi} \sin\theta\, d\theta}$$

$$= R \frac{\int_0^{\pi} \left[\frac{1-\cos 2\theta}{2}\right] d\theta}{[-\cos\theta]_0^{\pi}}$$

$$= \frac{R}{2} \frac{\left[\int_0^{\pi} d\theta - \int_0^{\pi} \cos 2\theta\, d\theta\right]}{[-\cos\theta]_0^{\pi}}$$

160%

$$= \frac{\frac{R}{2}\left[ [\theta]_0^{\pi} - \left[\frac{\sin 2\theta}{2}\right]_0^{\pi} \right]}{[-\cos\theta]_0^{\pi}}$$

$$= \frac{\frac{R}{2}\left[ \pi - \frac{1}{2}\left[ \cancel{\sin 2\pi} - \cancel{\sin 0} \right] \right]}{-[\cos\pi - \cos 0]}$$

$$-[-1-1]$$

$$\boxed{y_c = \frac{\pi R}{4}} \text{ Ans.}$$

**QUESTION 3**

2D Structure (Planar)

$dm = \sigma\, dA$

Semi disc of Mass = M & Radius = R.

$Y_c = \dfrac{\int y\, dm}{\int dm}$ → y coordinate of CoM of element

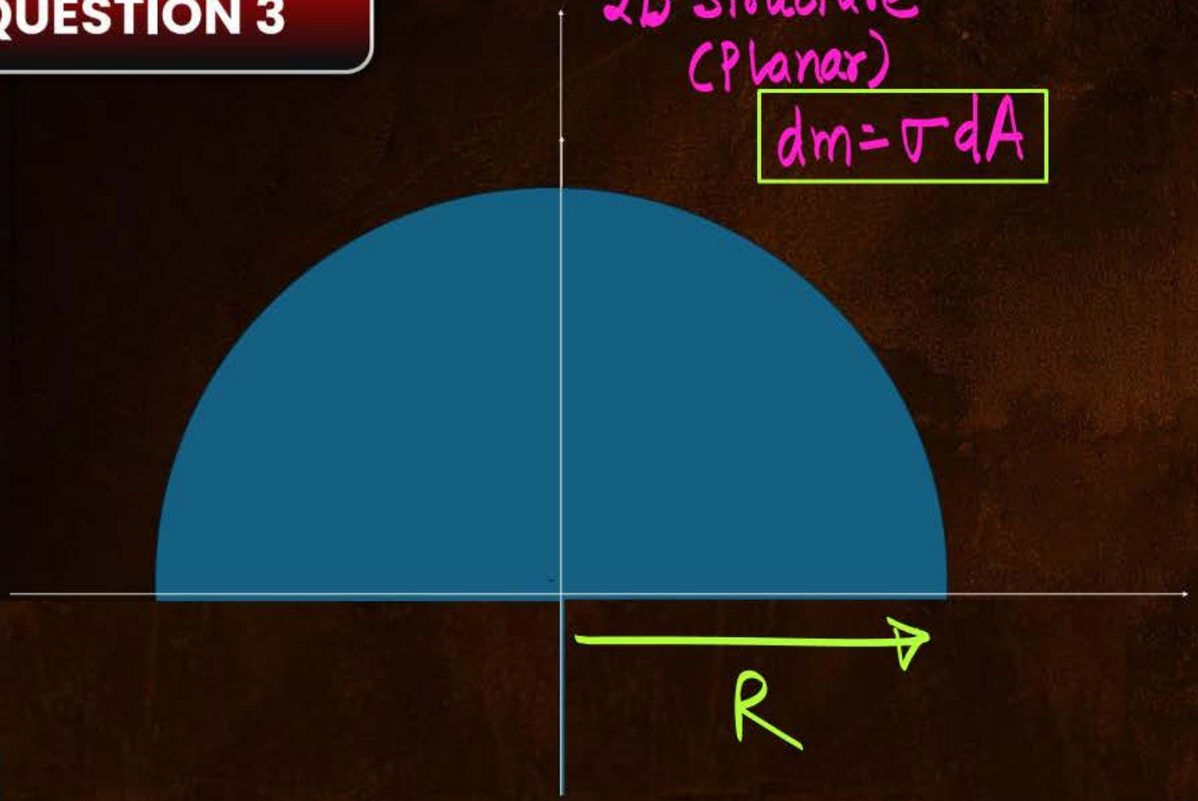
dm = mass of Element

$\dfrac{\pi R^2}{2} \rightarrow M$

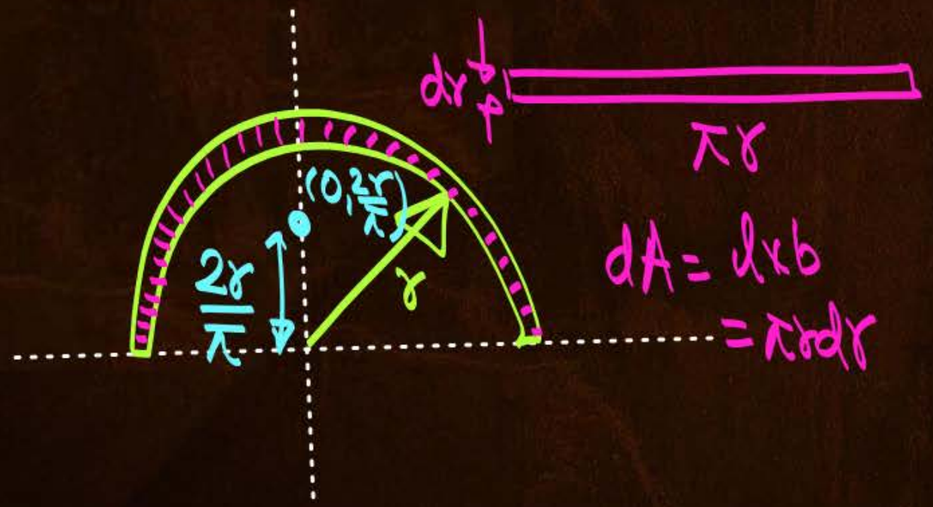
$1 \rightarrow \dfrac{M}{\frac{\pi R^2}{2}}$ (Areal Mass density) $\sigma$

$\pi r\, dr \rightarrow \dfrac{2M}{\pi R^2}\, \pi r\, dr$

$dm = \dfrac{2M r\, dr}{R^2}$




disc ⇒ Element = Ring.




$dA = l \times b$

$= \pi r\, dr$

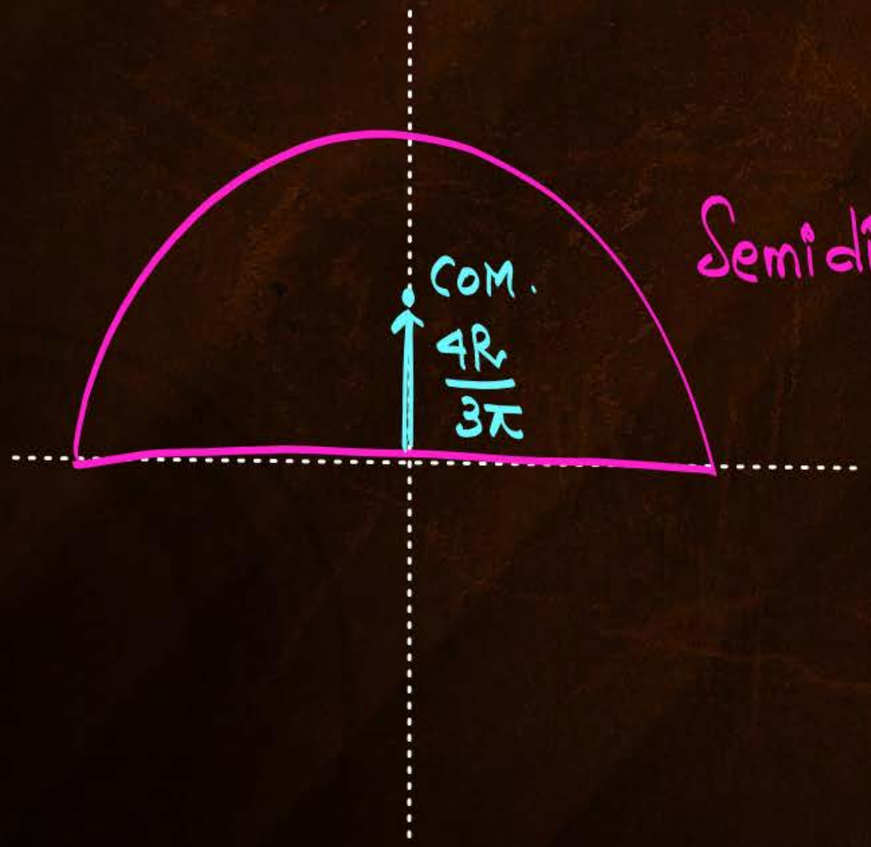
$$y_c = \frac{\int_0^R \frac{2r}{\pi}\,\frac{2Mr\,dr}{R^2}}{\int_0^R \frac{2Mr\,dr}{R^2}} \longrightarrow \text{Variable}$$

$$= \frac{\frac{4M}{\pi R^2}\int_0^R r^2\,dr}{\frac{2M}{R^2}\int_0^R r\,dr}$$

$$= \frac{2}{\pi}\,\frac{\left[\frac{r^3}{3}\right]_0^R}{\left[\frac{r^2}{2}\right]_0^R}$$

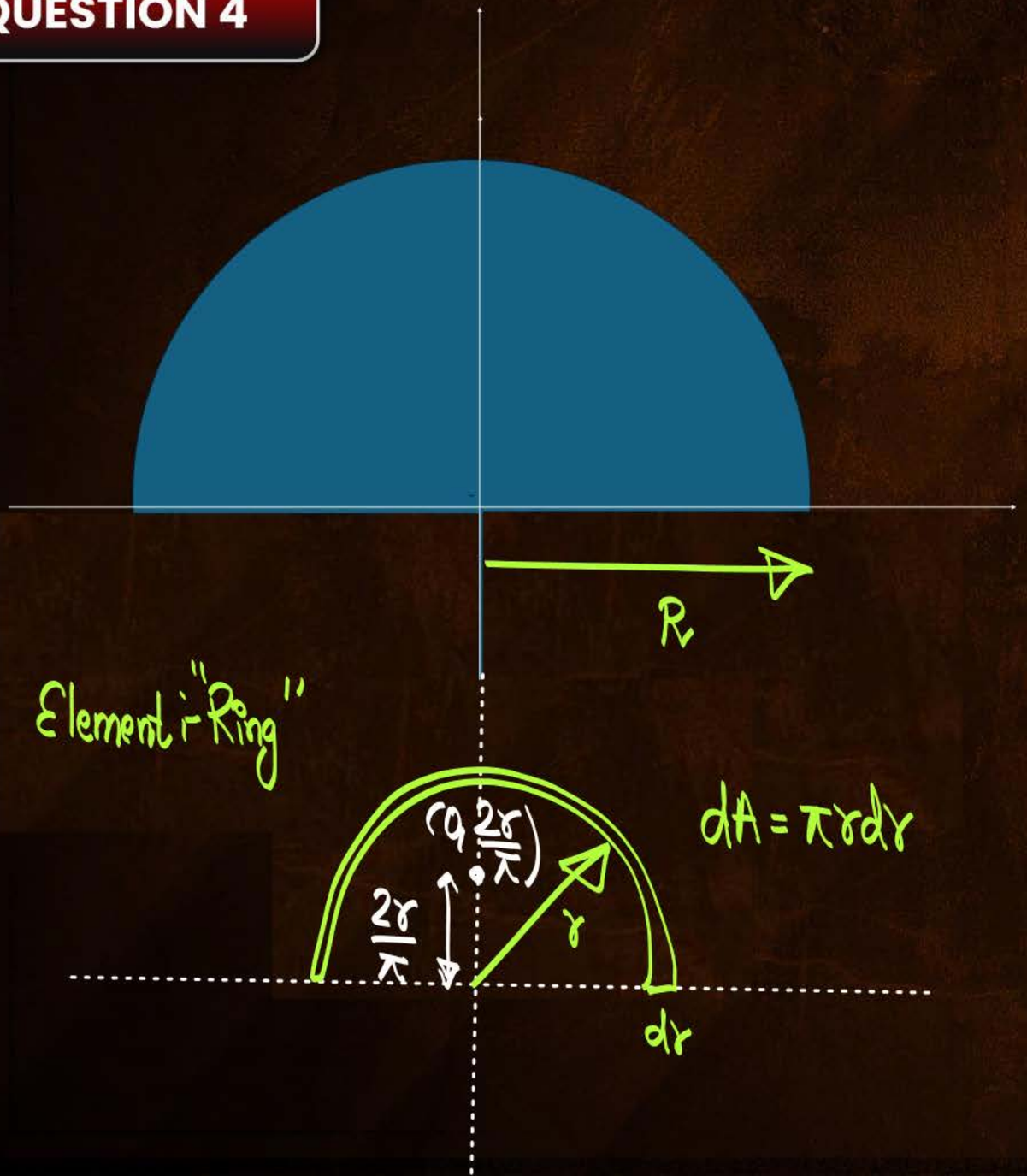
$$= \frac{2}{\pi} \times \frac{R^3}{3} \times \frac{2}{R^2}$$

$$= \frac{4R}{3\pi}$$




Semidisc

# QUESTION 4




Semi disc of Radius = R

has Non uniform Mass density.

$$\sigma = \sigma_0 r^2$$

where r is from origin.

$Y_c$ → y coord of COM

$$Y_c = \frac{\int y\, dm}{\int dm}$$

of Element $dm = \sigma dA$

$= \sigma_0 r^2 \pi r dr$

T → Variable $dm = \sigma_0 \pi r^3 dr$

$$= \frac{\int_0^R \frac{2r}{\pi} \times \sigma_0 \pi r^3 dr}{\int_0^R \sigma_0 \pi r^3 dr}$$

$$Y_c = \frac{\int_0^R \frac{2r}{\pi} \sigma_0 t r^3 dr}{\int_0^R \sigma_0 \pi r^3 dr} = \frac{2\sigma_0 \int_0^R r^4 dr}{\sigma_0 \pi \int_0^R r^3 dr}$$

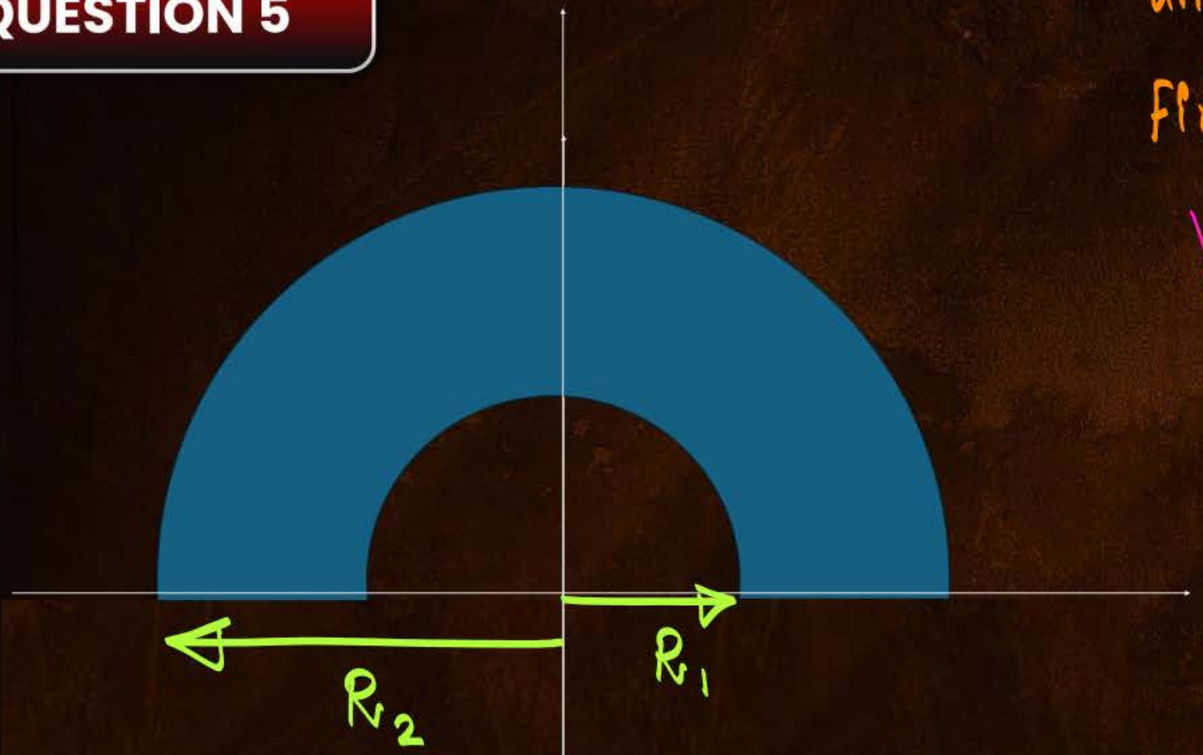
$$= \frac{2}{\pi} \frac{R^5}{5} \times \frac{4}{R^4}$$

$$= \frac{8R}{5\pi}$$

QUESTION 5

uniform annular disc of Mass density = $\sigma$

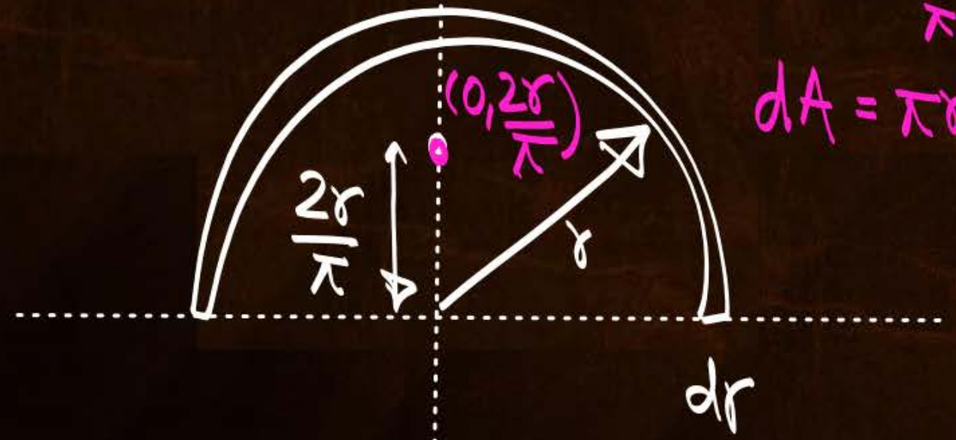
Find $Y_{com} = ?$




$y$ coord of COM of Element.

$$Y_{com} = \frac{\int y\,dm}{\int dm}$$

$$dm = \sigma\,dA = \sigma\,\pi r\,dr$$







$$dA = \pi r\,dr$$

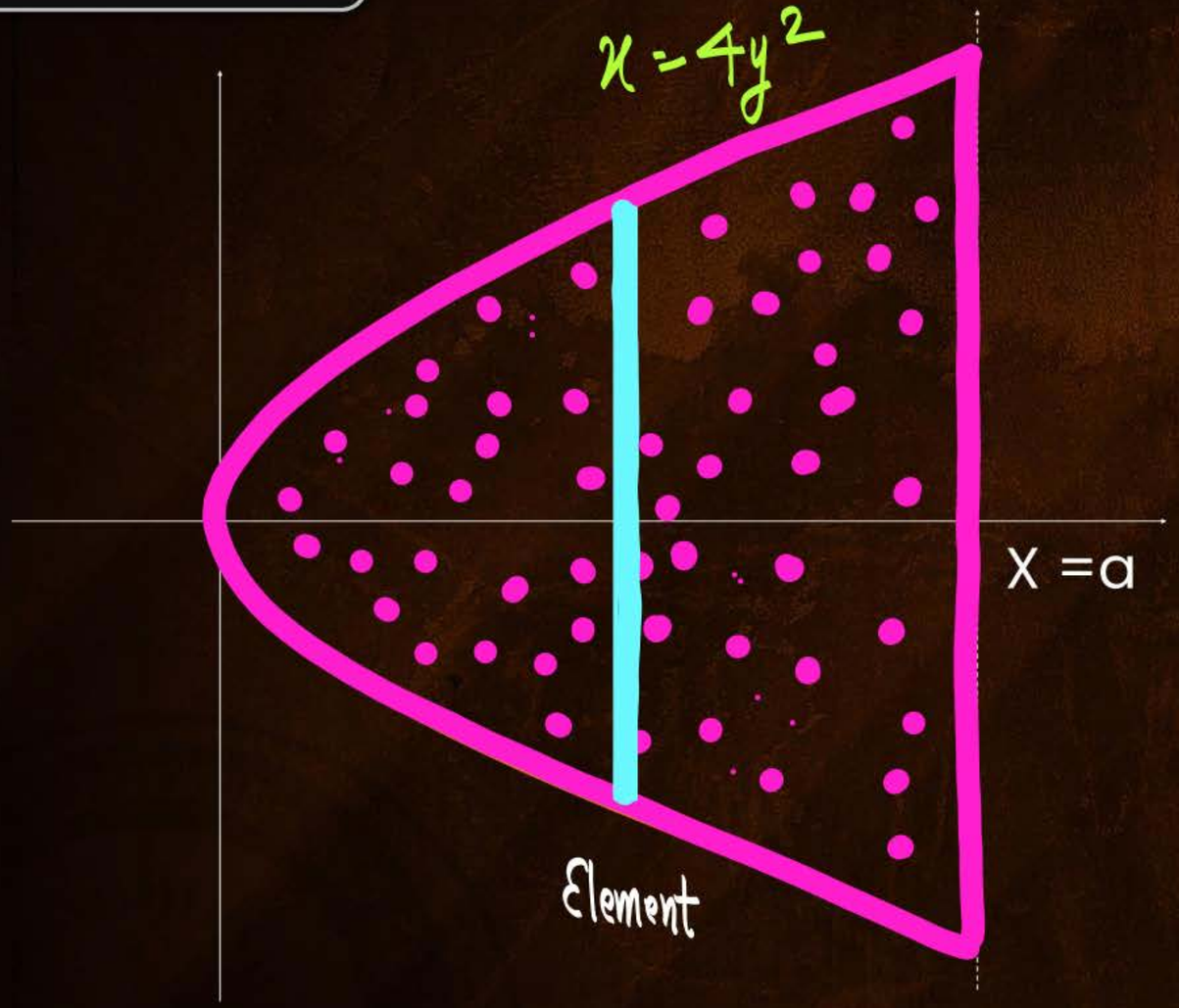
$$= \frac{\int_{R_1}^{R_2} \frac{2r}{\pi} \cdot \sigma\pi r\,dr}{\int_{R_1}^{R_2} \sigma\pi r\,dr}$$

→ Variable.

$$= \frac{2\sigma \int_{R_1}^{R_2} r^2\,dr}{\sigma\pi \int_{R_1}^{R_2} r\,dr}$$

$$Y_c = \frac{\frac{2}{\pi}\left[\frac{R_2^3}{3} - \frac{R_1^3}{3}\right]}{\left[\frac{R_2^2}{2} - \frac{R_1^2}{2}\right]}$$

QUESTION 6




# Parabolic sheet of uniform mass density $= \sigma$.

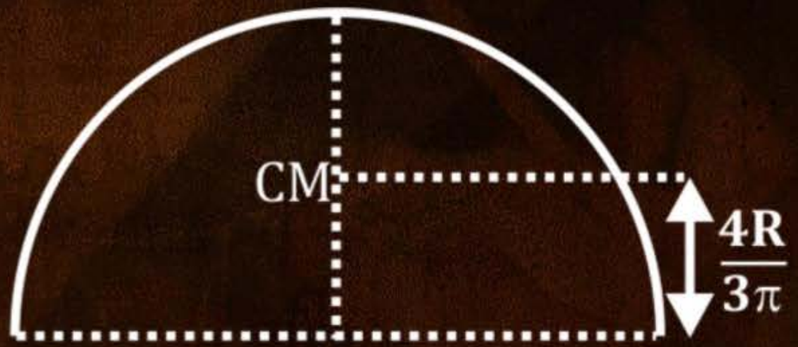
find $X_c = ?$

## ❖ Centre of mass of some uniform symmetric bodies are

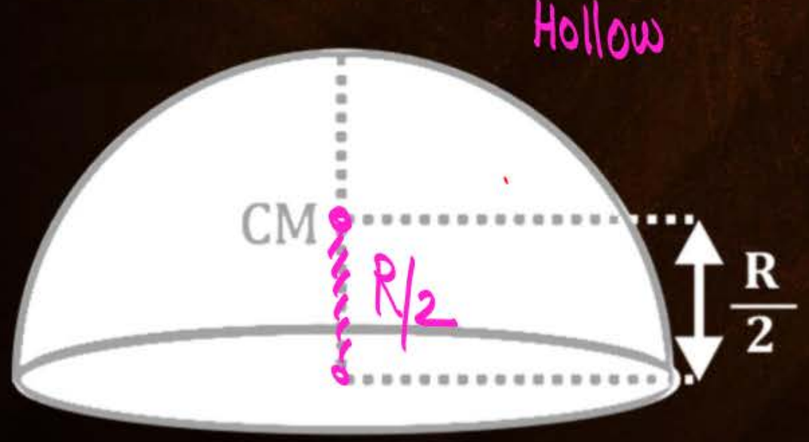
(i) Semicircular ring of radius R



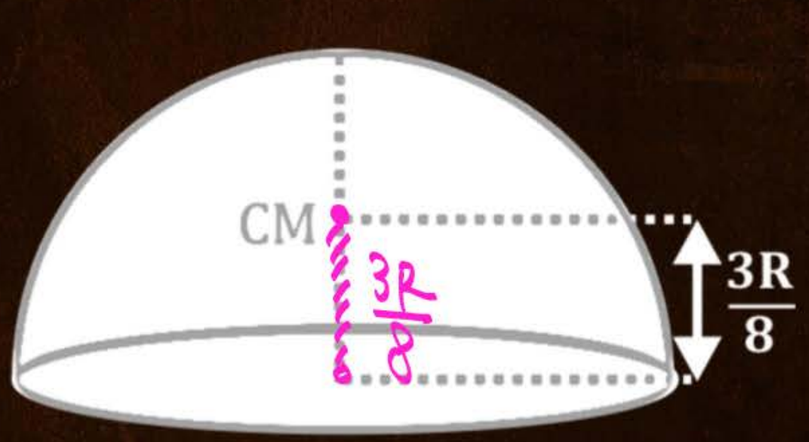

(ii) Semicircular disc




(iii) Hemispherical shell




(iv) Solid hemisphere

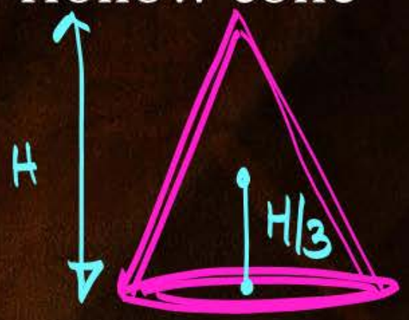


(v) Solid cone

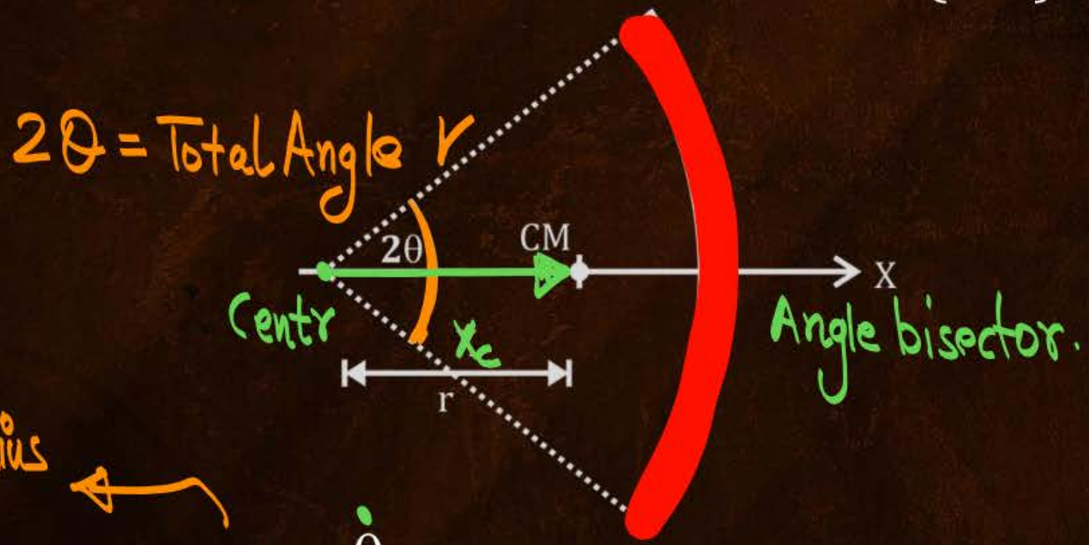



#

(vi) Hollow cone



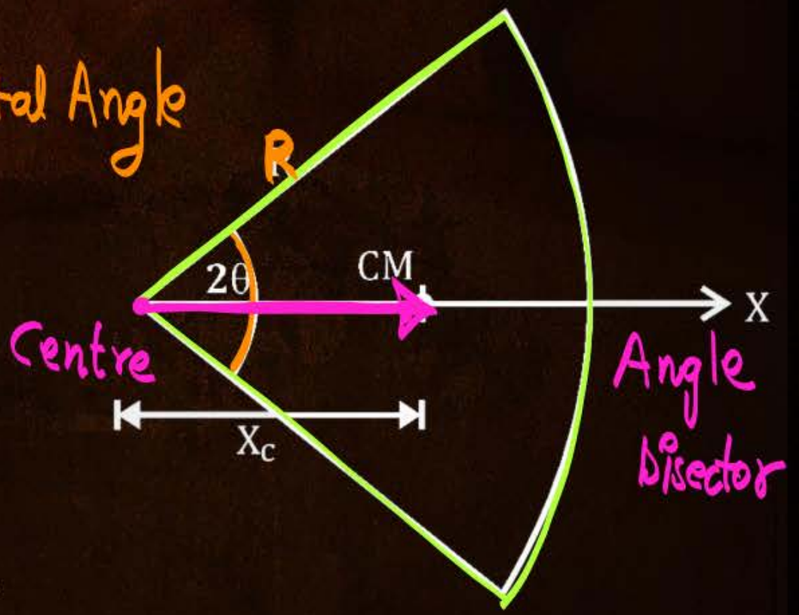

(vii) Circular Arc




$$x_c = \frac{r \sin\theta}{\theta}$$

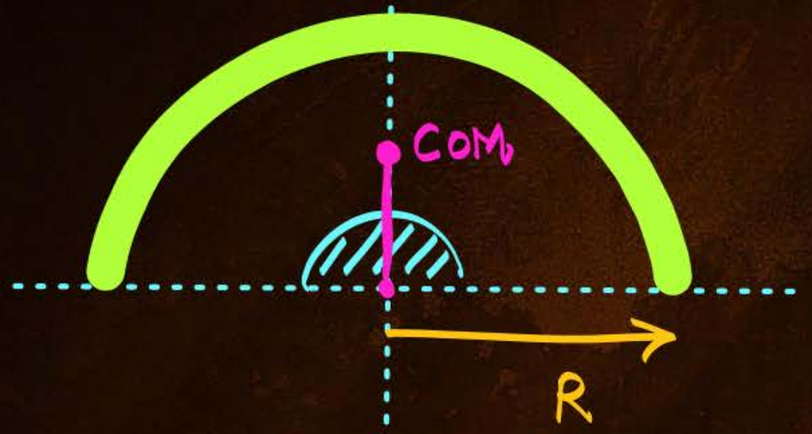
Note: Here θ is in radians.

(viii) Sector of a circular plate




$$x_c = \frac{2R \sin\theta}{3\theta}$$

Example.




Angle bisector.

general $\frac{r \sin\theta}{\theta}$

$$y = \frac{R \sin(\pi/2)}{\pi/2} = \frac{2R}{\pi}.$$

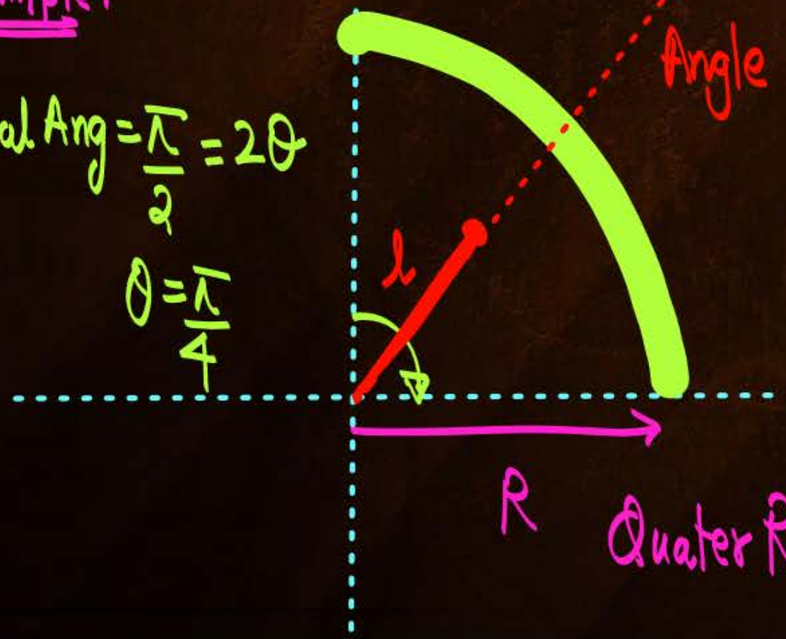
Total Angle $= \pi = 2\theta$

$\theta = \pi/2$

Example:-

Total Ang $= \frac{\pi}{2} = 2\theta$

$\theta = \frac{\pi}{4}$




$$l = \frac{R \sin(\pi/4)}{\pi/4} = \frac{4R}{\pi} \times \frac{1}{\sqrt{2}} = \frac{2\sqrt{2}R}{\pi}.$$

# # disc

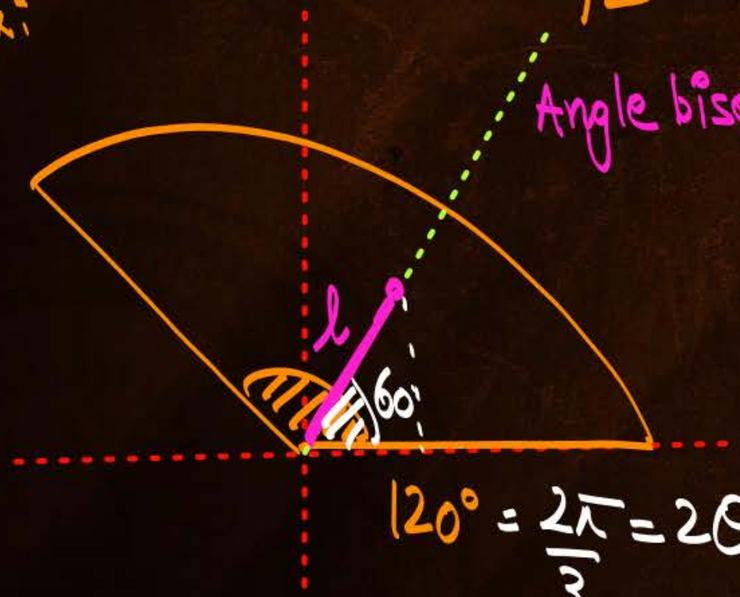



Semidisc $\quad l = \dfrac{2R\sin\theta}{3\theta} = \dfrac{2R\sin(\pi/2)}{3(\pi/2)} = \dfrac{4R}{3\pi}$.

Total Angle $= \pi = 2\theta$

$\theta = \pi/2$

Ex:




$120° = \dfrac{2\pi}{3} = 2\theta$

$\theta = \pi/3$

find Coordinate of COM :-

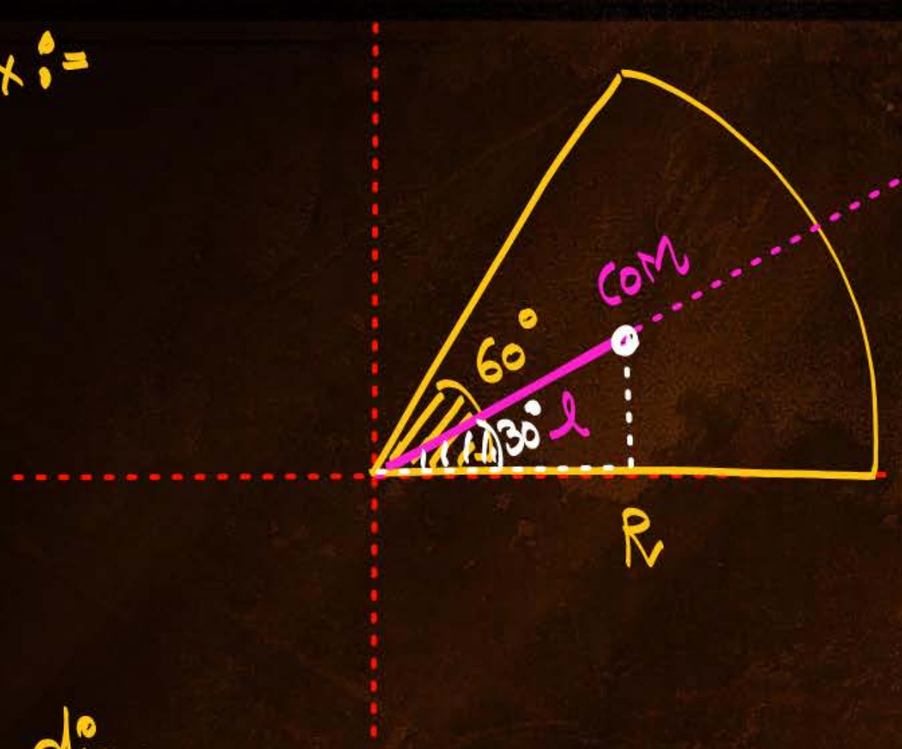
$l = \dfrac{2R\sin\pi/3}{3(\pi/3)} = \dfrac{R\sqrt{3}}{\pi}$

$= l\cos 60, l\sin 60$.

$= \dfrac{R\sqrt{3}}{\pi}\times\dfrac{1}{2}, \dfrac{R\sqrt{3}}{\pi}\times\dfrac{\sqrt{3}}{2}$

$= \left(\dfrac{R\sqrt{3}}{2\pi}, \dfrac{3R}{2\pi}\right)$.

Ex :-




Angle bisector

Coordinate of COM $= l\cos 30, l\sin 30$

$= \frac{2R}{\pi}\frac{\sqrt{3}}{2}, \frac{2R}{\pi}\times\frac{1}{2}$

$= \left(\frac{R\sqrt{3}}{\pi}, \frac{R}{\pi}\right)$.

Total Angle $= 2\theta = \frac{\pi}{3}$

$\theta = \frac{\pi}{6}$

disc

$$l = \frac{2R\sin\theta}{3\theta}$$

$$= \frac{2R\sin \pi/6}{3(\pi/6)} = \frac{2R}{\pi}$$

# Additive / Negative System

## In terms of Mass

Add $X_c = \frac{m_1x_1 + m_2x_2 + \cdots}{m_1 + m_2 + \cdots}$

$Y_c = \frac{m_1y_1 + m_2y_2 + \cdots}{m_1 + m_2 + \cdots}$

Neg ⇒ (Removal)

$X_c = \frac{m_1x_1 - m_2x_2 - m_3x_3 \cdots}{m_1 - m_2 - m_3 \cdots}$

$Y_c = \frac{m_1y_1 - m_2y_2 - m_3y_3 \cdots}{m_1 - m_2 - m_3 \cdots}$

## 2D objects (In terms of Area)

Add $X_c = \frac{A_1x_1 + A_2x_2}{A_1 + A_2}$

$Y_c = \frac{A_1y_1 + A_2y_2}{A_1 + A_2}$

Neg $= X_c = \frac{A_1x_1 - A_2x_2}{A_1 - A_2}$

$Y_c = \frac{A_1y_1 - A_2y_2}{A_1 - A_2}$

## 3D (In terms of Volume)

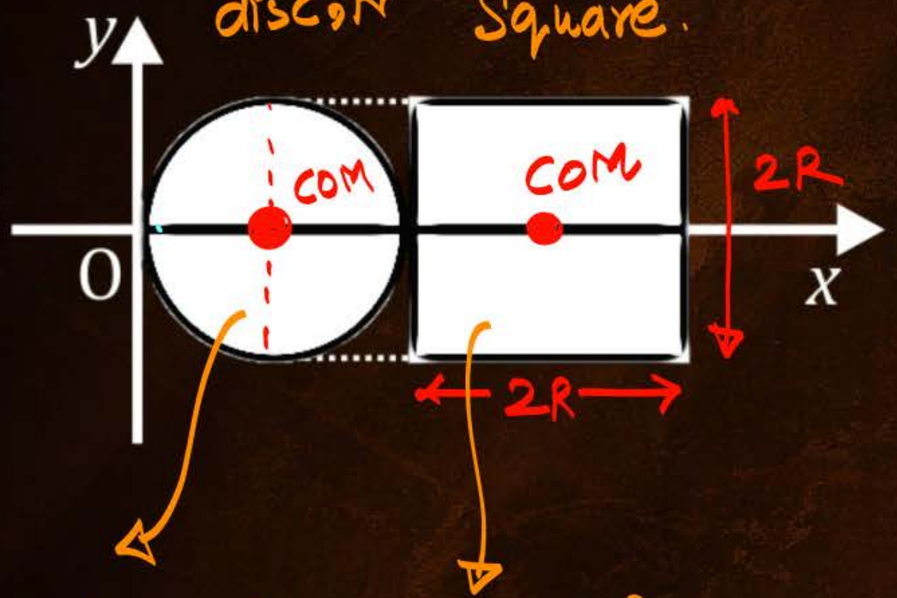
Add $X_c = \frac{V_1x_1 + V_2x_2}{V_1 + V_2}$

$Y_c = \frac{V_1y_1 + V_2y_2}{V_1 + V_2}$

Neg $X_c = \frac{V_1x_1 - V_2x_2}{V_1 - V_2}$

$Y_c = \frac{V_1y_1 - V_2y_2}{V_1 - V_2}$

**QUESTION 9** **Centre of mass of composite bodies:**




disc, R  Square.

$A_1 = \pi R^2$ $(R, 0)$

$A_2 = 4R^2$ $(3R, 0)$

"Add"

$$X_c = \frac{A_1x_1 + A_2x_2}{A_1 + A_2}$$

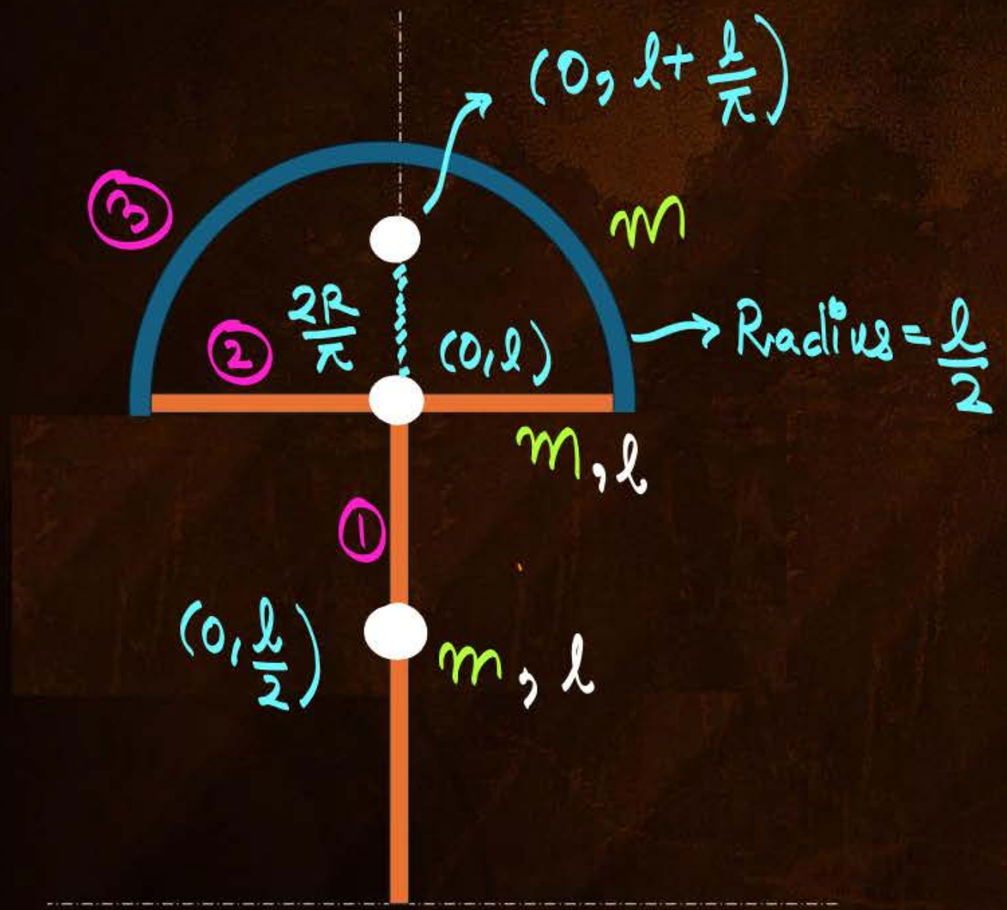
$$= \frac{\pi R^2(R) + 4R^2(3R)}{\pi R^2 + 4R^2}$$

$$= \frac{\pi R^3 + 12R^3}{\pi R^2 + 4R^2} = R\left[\frac{12 + \pi}{4 + \pi}\right]$$

$Y_c = 0$

$\therefore$

# QUESTION 10




"Add"

$$Y_c = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1 + m_2 + m_3}$$

① $m$ $(0, \frac{l}{2})$
② $m$ $(0, l)$
③ $m$ $(0, l + \frac{l}{\pi})$

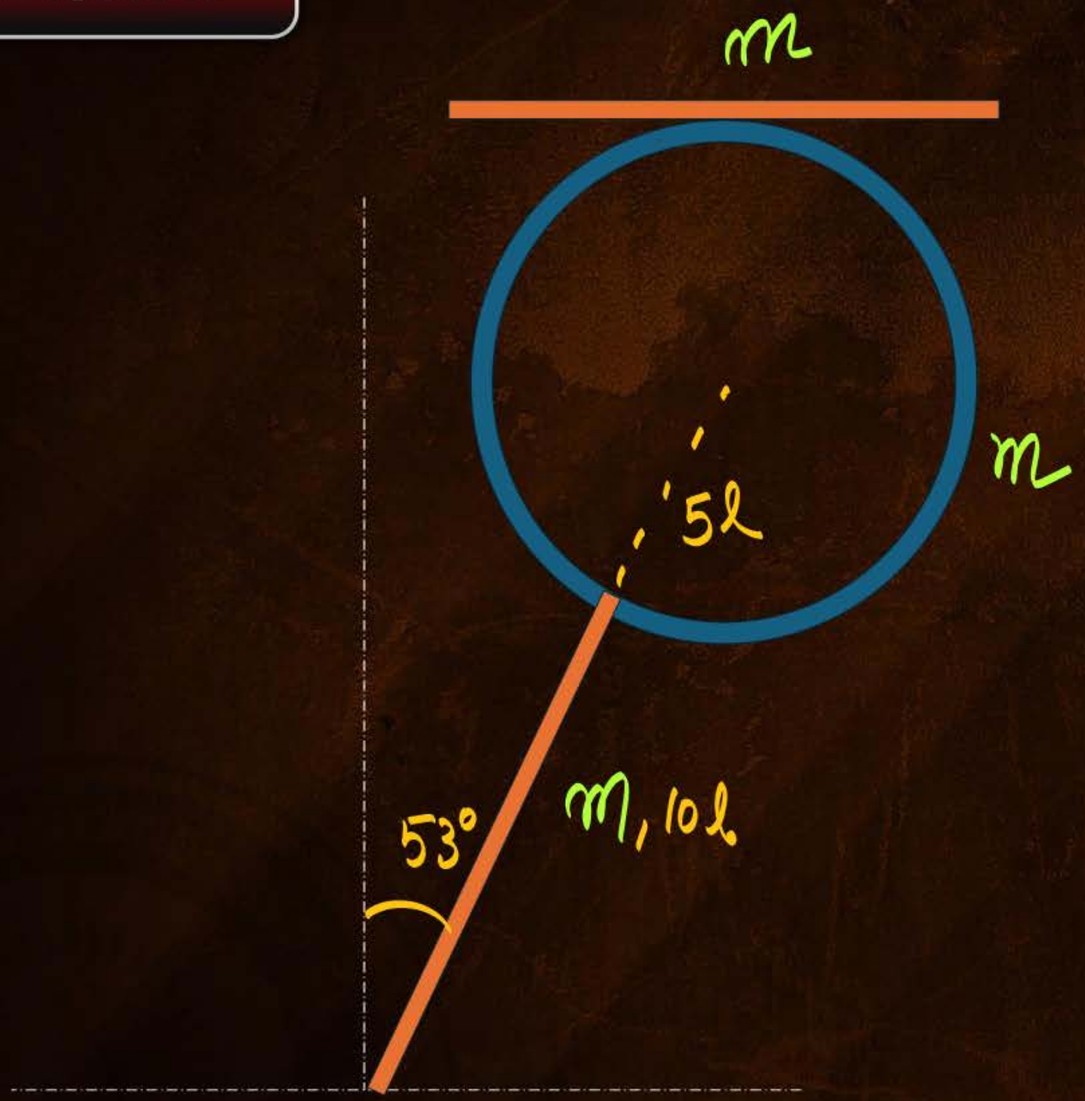
$$= \frac{m\left(\frac{l}{2}\right) + m(l) + m\left(l + \frac{l}{\pi}\right)}{3m}$$

$$= \frac{\frac{3ml}{2} + ml\left(1 + \frac{1}{\pi}\right)}{3m}$$

$$Y_c = \frac{l}{2} + \frac{l}{3}\left(1 + \frac{1}{\pi}\right)$$

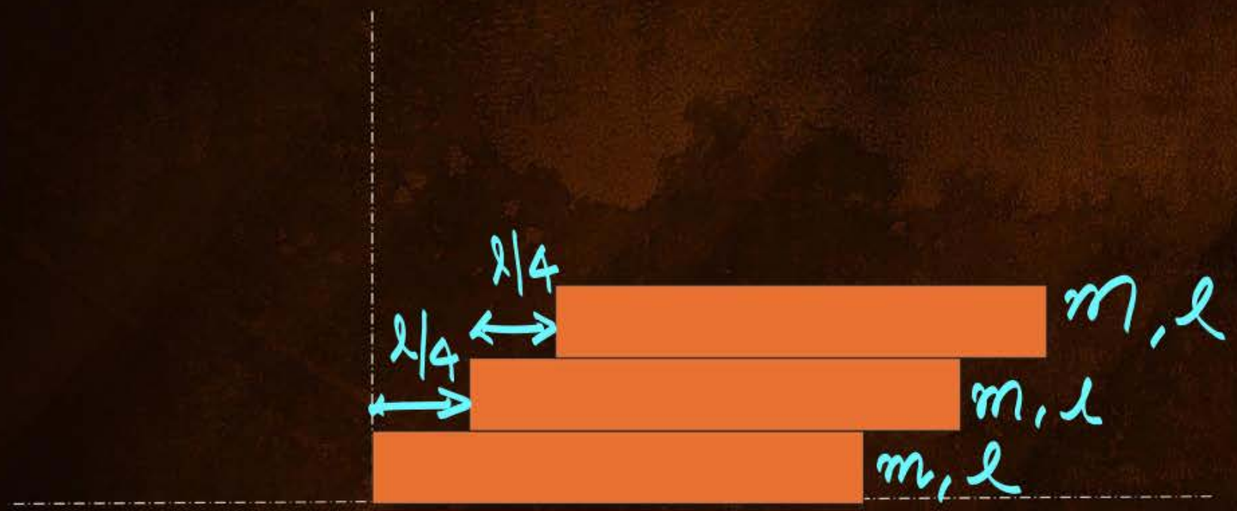
$$X_c = 0$$

QUESTION 11



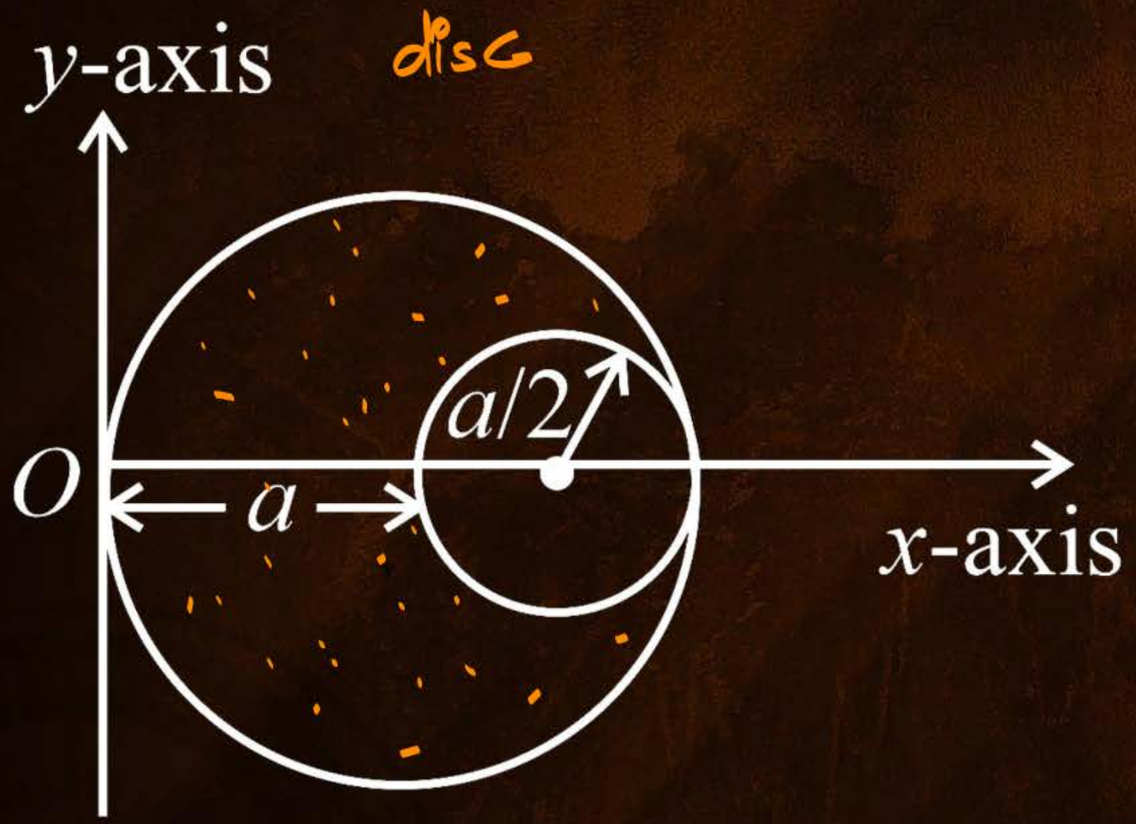
QUESTION 12

3 Rods

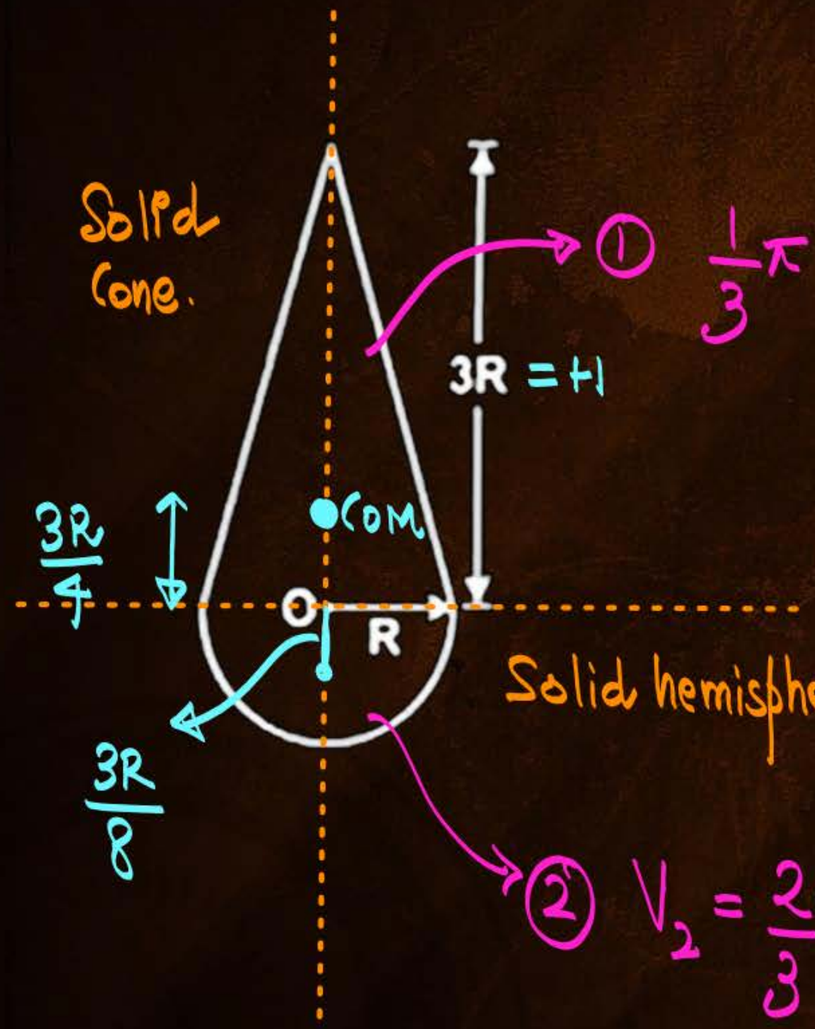



$X_c = ?$

QUESTION 13



# QUESTION 14



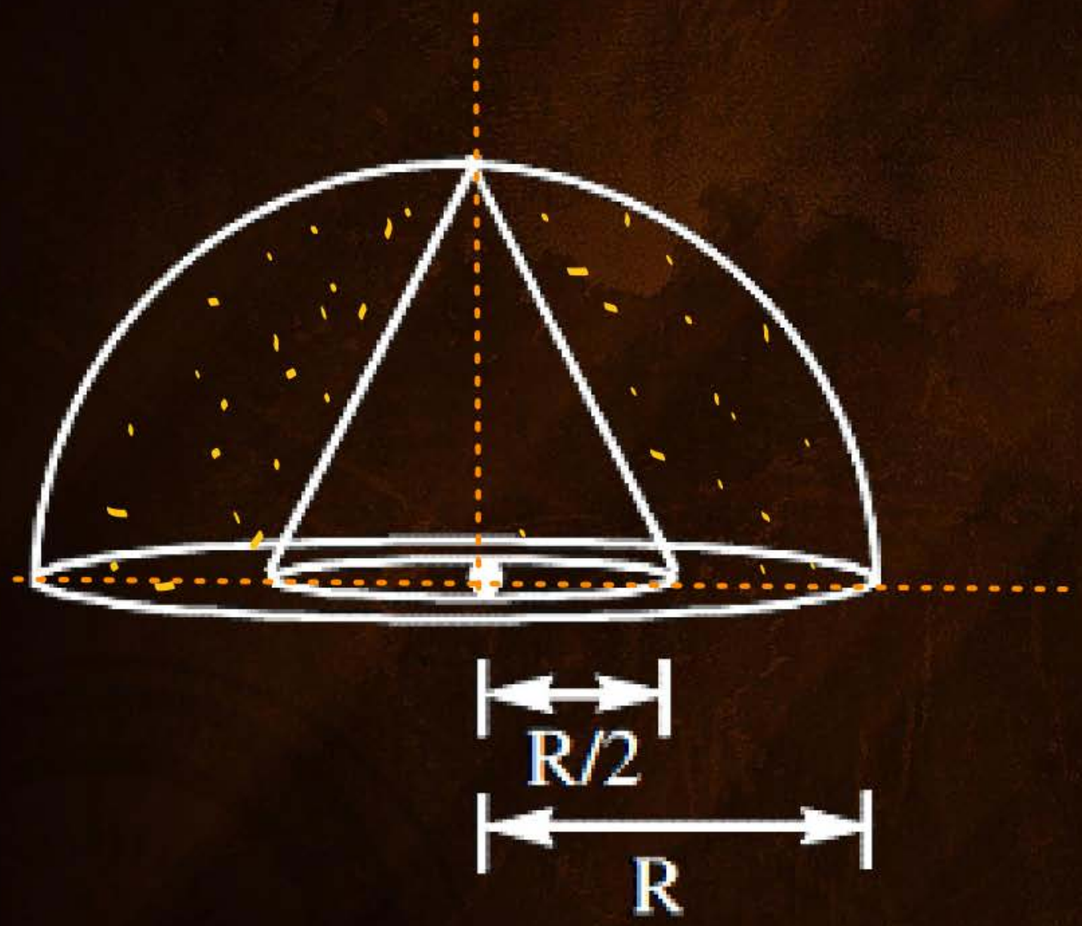

"Add" $Y_c = \frac{V_1 y_1 + V_2 y_2}{V_1 + V_2} = \frac{\pi R^3\left(\frac{3R}{4}\right) + \frac{2}{3}\pi R^3\left(\frac{-3R}{8}\right)}{\pi R^3 + \frac{2}{3}\pi R^3}$

① $\frac{1}{3}\pi R^2 H = \frac{1}{3}\pi R^2(3R) = \pi R^3$ $\left(0, \frac{3R}{4}\right)$

② $V_2 = \frac{2}{3}\pi R^3$ $\left(0, -\frac{3R}{8}\right)$

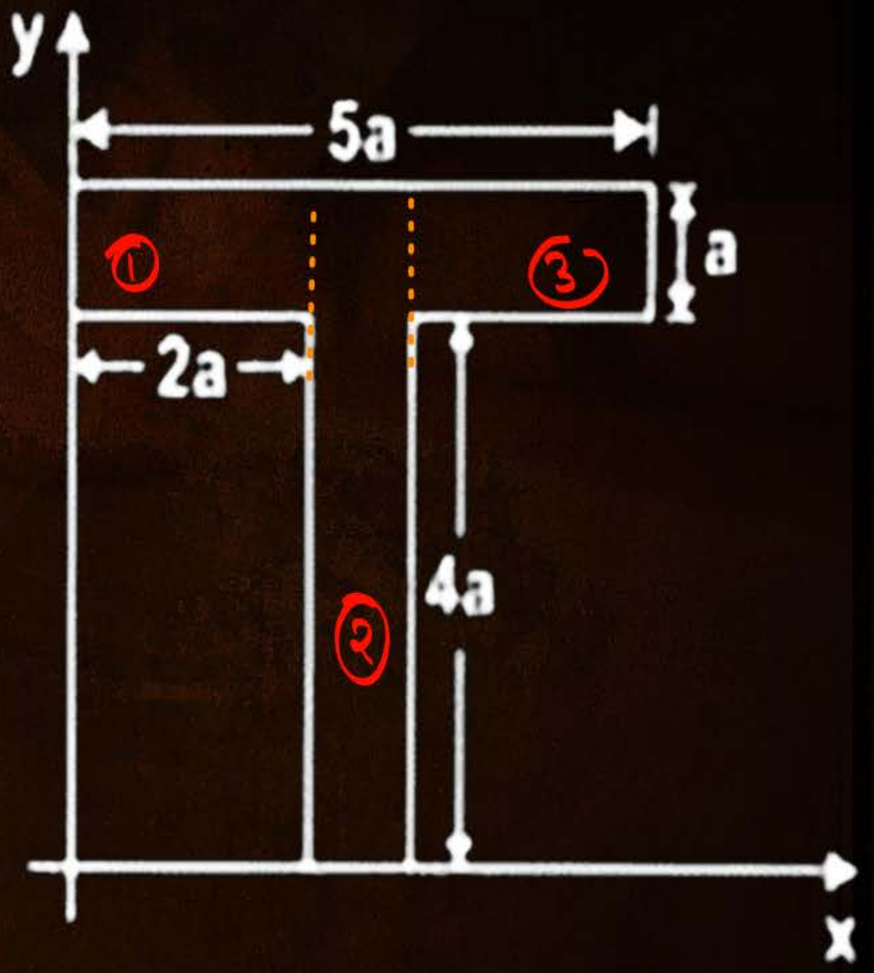
QUESTION 15

Cone is Removed.



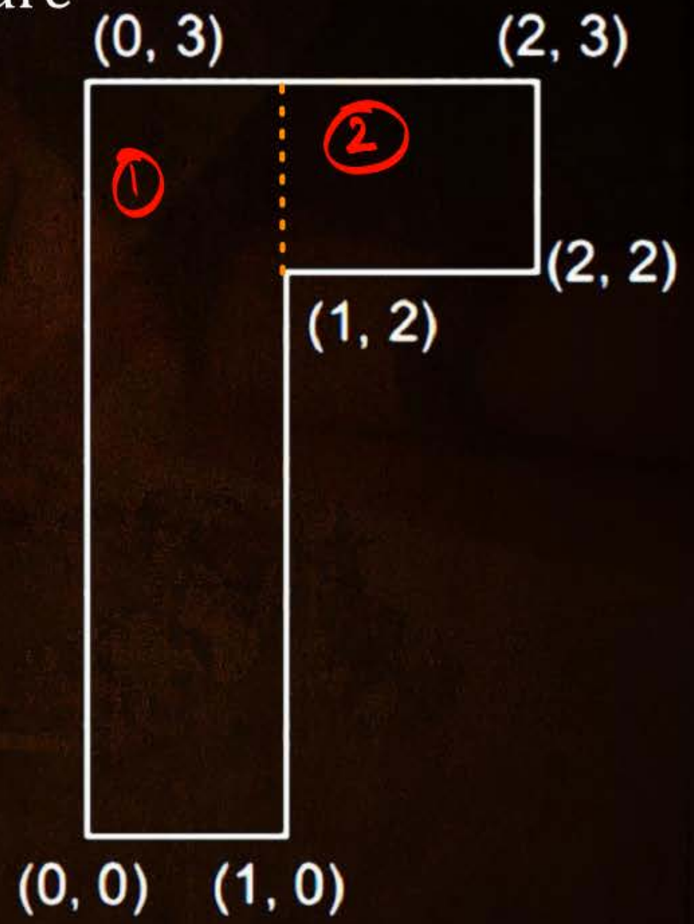
## QUESTION 16

Figure shows a uniform two dimensional T-shaped structure. Find co-ordinates of centre of mass of this object.



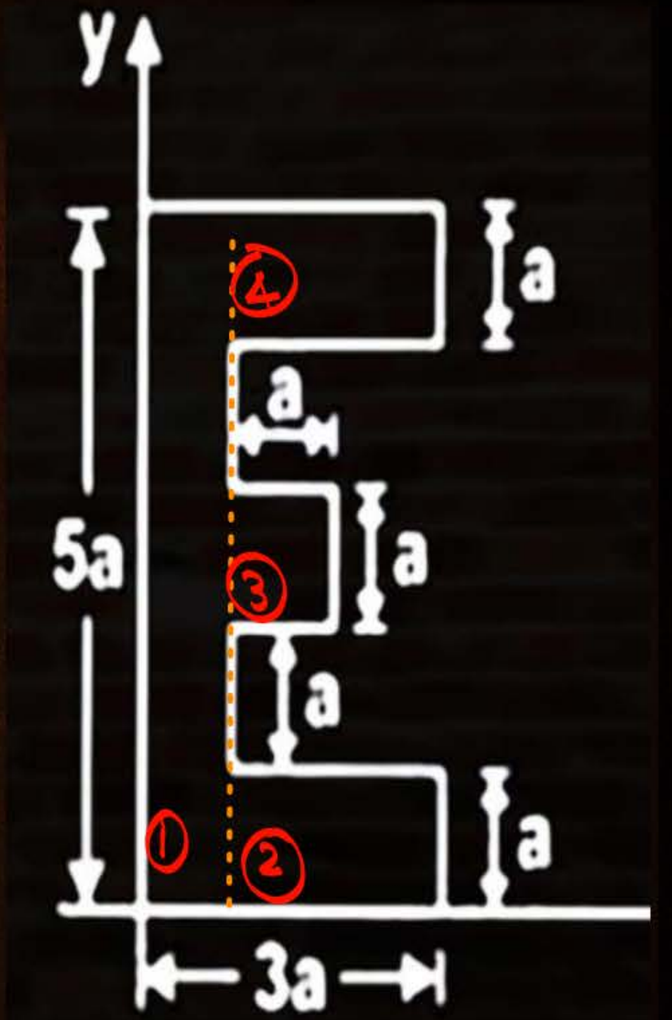
## QUESTION 17

Find the coordinates of centre of mass of the lamina, shown in figure



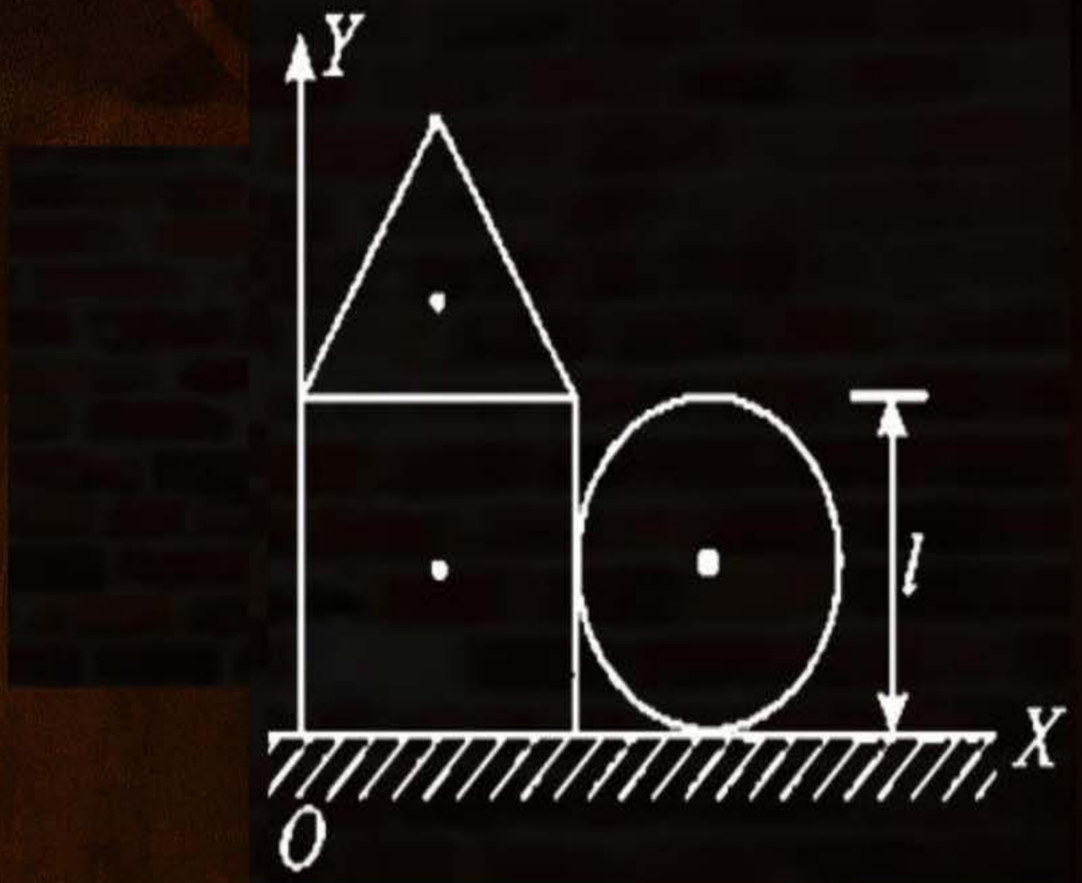
## QUESTION 18

Figure shows a uniform plate oi shape $E$ as shown. Find the co-ordinates of centre of mass of this structure.



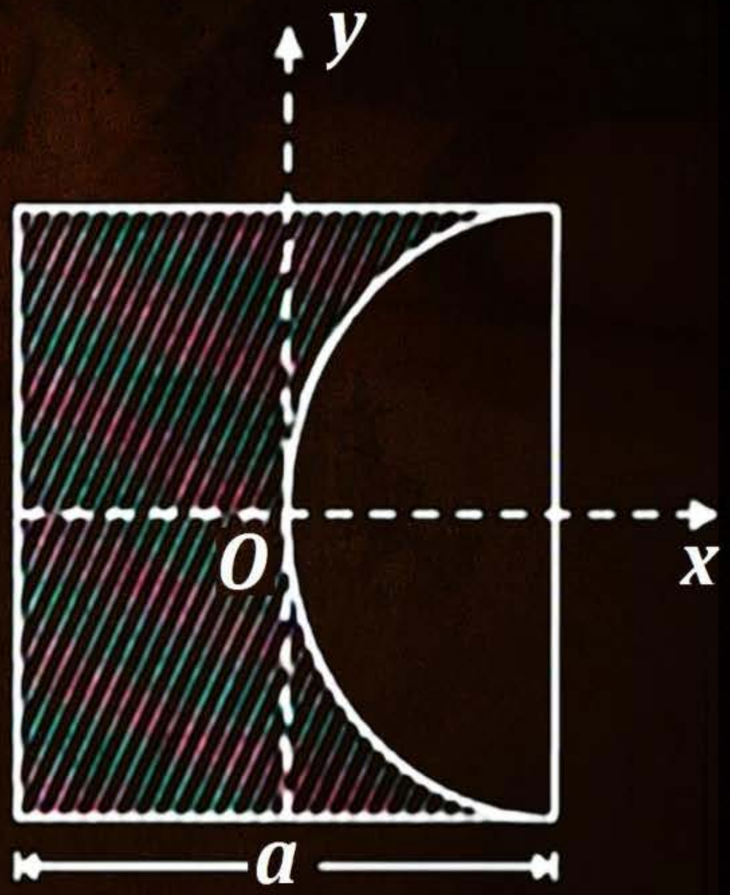
## QUESTION 19

Three laminar objects of uniform density a square, a disc and an equilateral triangle are placed as shown in figure . Find the coordinates of center of mass of the three bodies.



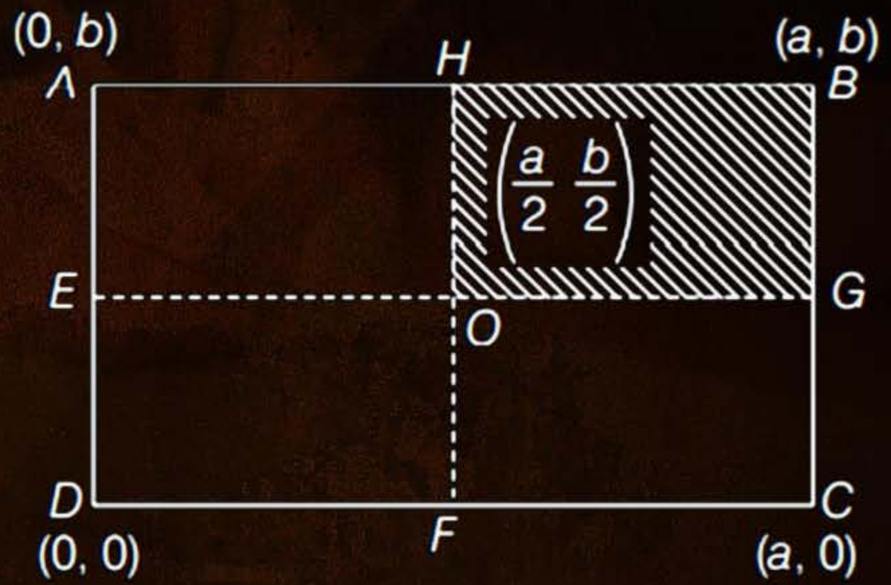
## QUESTION 20

Figure shows a square plate of side $a$ from which a semicircular disc is removed as shown. Find the centre of mass of the remaining object.



## QUESTION 21

A uniform rectangular thin sheet *ABCD* of mass M has length a and breadth b, as shown in the figure. If the shaded portion *HBGO* is cut-off, the coordinates of the centre of mass of the remaining portion will be




**A** $\left(\frac{2a}{3}, \frac{2b}{3}\right)$

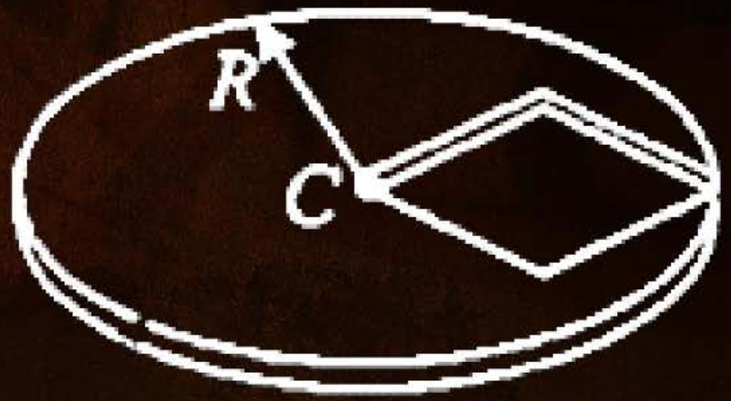
**B** $\left(\frac{5a}{12}, \frac{5b}{12}\right)$

**C** $\left(\frac{3a}{4}, \frac{3b}{4}\right)$

**D** $\left(\frac{5a}{3}, \frac{5b}{3}\right)$

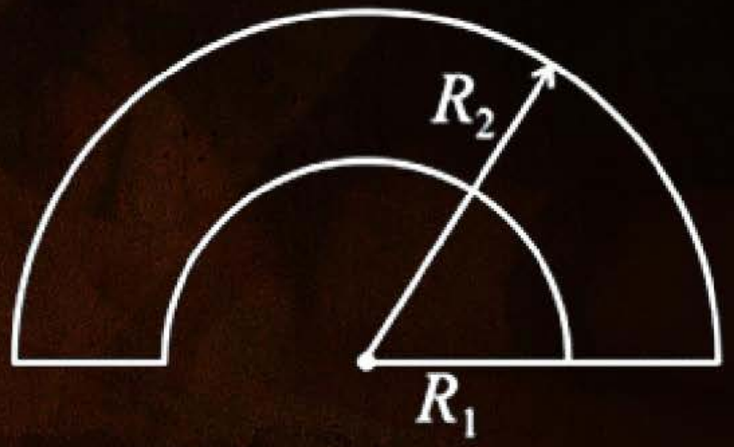
## QUESTION 22

Find the centre of mass of the disc shown in figure. A square is removed from the disc as shown .



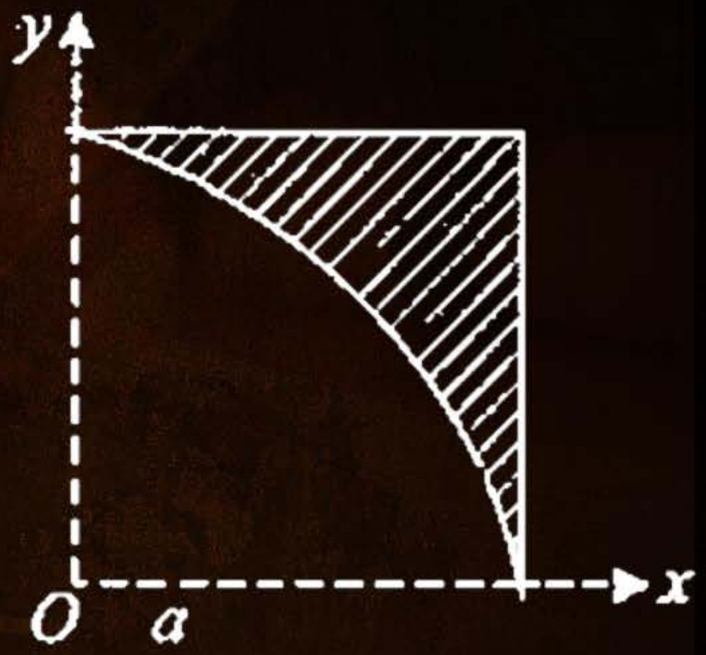
## QUESTION 23

Find the centre of mass of an annular half disc shown in figure.



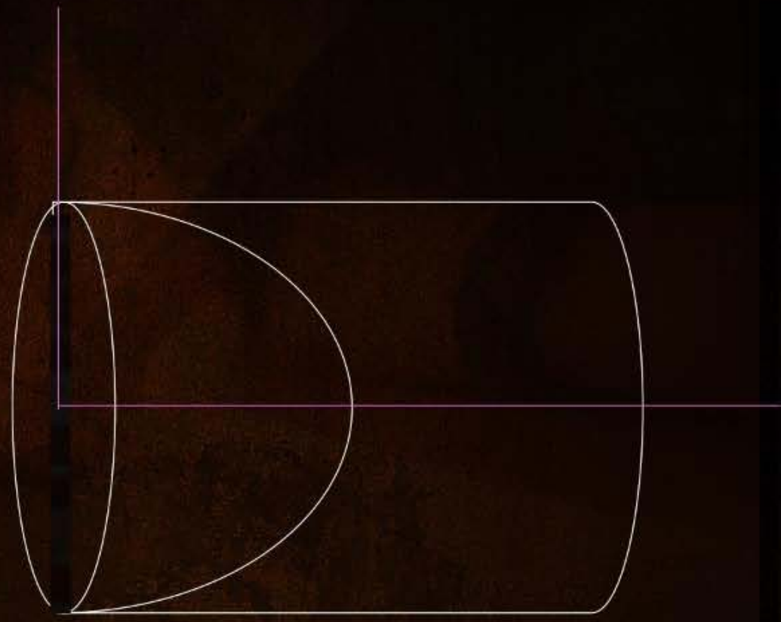
## QUESTION 24

From a square plate of side $a$, $a$ quarter circular disc of radius $a$ is removed as shown in figure. Find the coordinates of center of mass of the remaining part.



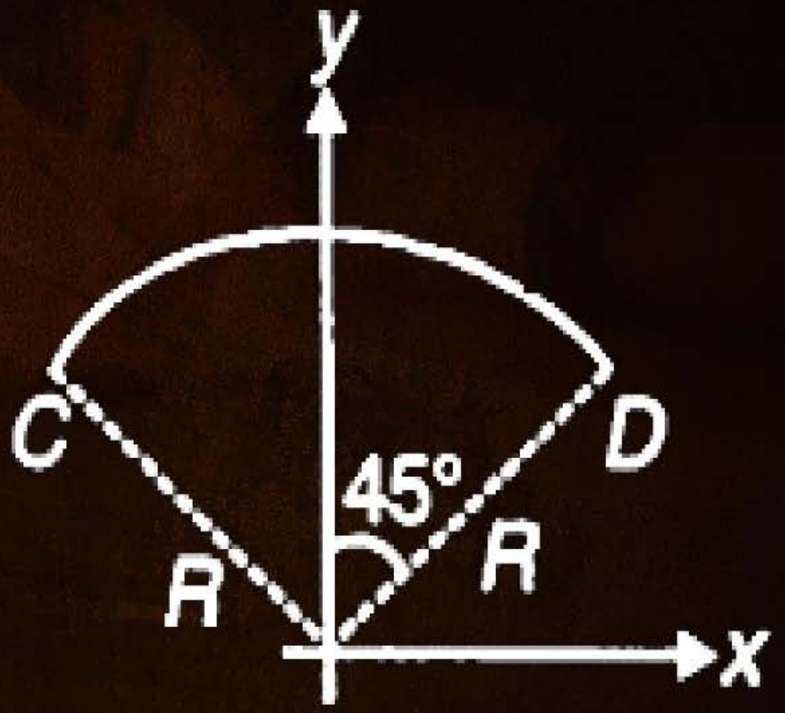
## QUESTION 25

From a cylinder of length 2R and radius R, a hemispherical portion is removed, find COM of remaining portion.



## QUESTION 26

A circular arc $CD$ of thin wire frame of radius $R$ and mass $M$ with center at origin makes an angle of $90°$ at the origin. The $y$-coordinate center of mass of the arc lies at




**A** $\frac{\sqrt{2}R}{\pi}$

**B** $\frac{R}{2\pi}$

**C** $\frac{2R}{\pi}$

**D** $\frac{2\sqrt{2}R}{\pi}$



# Homework

- Complete Questions from module till topic of finding COM.

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